

Improving campus dining operations using capacity and queue management: A simulation-based case study

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ABSTRACT

With growing consumer demand, dining facilities at higher education institutions have started to provide a wide assortment of food products catering to the needs of every individual. In developed countries like the United States, labor is an expensive asset, and campus dining facilities are struggling to utilize their employees efficiently and achieve service excellence. This research is motivated by a real-life case study of a campus dining service (CDS), which is an all-you-can-eat establishment with an annual influx of 500,000 students. The CDS has eight food stations but experiences long queues only at individual stations that prepare made-to-order (MTO) meals, thereby leading to excessive wait at the facility. Given the characteristics of the CDS, the objective of this paper is to evaluate the impact of capacity reallocation and queue management on customer waiting time. The methodology integrates the six sigma DMAIC (Define, Measure, Analyze, Improve, and Control) framework with a discrete event simulation modeling approach to understand, analyze, and improve the service responsiveness at CDS. Five different alternatives pertaining to capacity reallocation and queue management are proposed. The experimental analyses indicate that reallocating underutilized workers to high demand MTO stations can reduce the average waiting time by 29%. Besides, when adopting a combination of server reallocation and queue management, the CDS can curtail wait times by up to 45%. Finally, the alternatives are evaluated with respect to multiple criteria (cost and ease of implementation, responsiveness, and sustainability) to provide managerial insights to the CDS.

1. Introduction

Campus Dining Service (CDS) is an integral part of the college environment and is perceived as a critical determinant for selecting a university by 1 in 4 students (Technomic, 2019). As a result, CDS facilities are striving to provide competitive pricing and offer large varieties by including made-to-order (MTO) foods, multi-cuisine options, dietary accommodations, and buffet-style dining (El-Said & Fathy, 2015; Joung, Lee, Kim, & Huffman, 2014). Nevertheless, over two-thirds of the students are discontented with their university's food service, and 69% opt to eat at nearby restaurants at least once a week (Technomic, 2019). This is mainly because of the long queues and excessive waiting time for service at the CDS, especially during peak-hours (Klassen et al., 2005; Tang, Zhang, Zhang, & Wang, 2019).

One of the critical dimensions of excellent service quality is *responsiveness* – the ability to provide prompt service with minimal waits (Parasuraman, Zeithaml, & Berry, 1988). While expanding the

workforce is known to increase responsiveness (Iqbal, Whitman, & Malzahn, 2012; Ma, Beeker, McCloskey, Moldawer, & Williams, 2012; Velumani, Pitchiah, & Tang, 2017), most CDS are struggling to attract and retain full-time staff as they only offer temporary employment. The hiring of student workers is a viable alternative to this problem. However, students typically prefer to acquire professional experience that is directly pertinent to their field of study (Gurgol, 2019). As a result, understaffing is a primary cause for longer waiting times at these locations (McCarthy, 2018). Thus, to attenuate service delays, it is critical to efficiently allocate the limited labor resources under demand uncertainty as students arrive in batches spread across unequal intervals of time (Curin, Vosko, Chan, & Tsimhoni, 2005).

This research is motivated by a real-life case study of a CDS facility that is owned and operated by a university in mid-Missouri, which has buffet lines of four MTO stations, three pre-cooked stations, and one beverage station. The dining service currently experiences an average waiting time of over 17 min at individual MTO stations. Further, a

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recent survey of over 500 students visiting the CDS location under study showed that 32% are unhappy with their wait time experiences, and over 15% opted to eat at nearby privately-owned restaurants. Recognizing this potential opportunity to improve service quality and sales revenue, this paper aims to examine the influence of capacity and queue management strategies on responsiveness. In particular, this study investigates the impact of the following CDS modifications on customer waiting time - (1) shifting existing workforce towards labor-intensive tasks, (2) untethering customers from queue upon placing an order, and (3) the combined effect of reallocating workers and untethering customers from the queue.

2. Literature review

Previous work have studied different business aspects of dining service, such as corporate social responsibility (Lee, Singal, & Kang, 2013), service quality (DiPietro, Parsa, & Gregory, 2011), marketing strategies (Ghouri, Khan, Malik, & Razzaq, 2011), customer gratification (Gupta, McLaughlin, & Gomez, 2007), and employee turnover (Brandmeir & Baloglu, 2005). The present study mainly focuses on capacity and queue management strategies to reduce customer waiting time as these are found to be critical drivers of operational efficiency (Fragnière et al., 2010; Ghaleb, Suryahatmaja, & Alharkan, 2015; Zhao, 2016).

2.1. Role of capacity management on service responsiveness

Capacity management is an effective strategy utilized to solve complex problems in the hospitality and tourism industry (Pullman & Rodgers, 2010). Dining facilities, such as restaurants and CDS locations, carry a high labor cost (Kim, Jang & Shawn, 2020). In such cases, capacity decisions are made from various strategic perspectives, since labor costs affect profitability (Adenso-Díaz, González-Torre, & García, 2002). Most of the prior research on improving service responsiveness using capacity management has focused on expanding the workforce (Curin et al., 2005; Ghaleb et al., 2015; Iqbal et al., 2012). Curin et al. (2005) conducted a simulation study of a CDS location to estimate the staffing level required to minimize waiting time and identified the need for an additional worker to achieve a 2-min reduction in waiting time per customer. A similar study was conducted by Iqbal et al. (2012) at another facility, where they identified the need for another staff to reduce the waiting time during peak hour operations significantly. On the contrary, Ghaleb et al. (2015) investigated the impact of removing redundant servers and combining queues on the waiting time. They concluded that a combination of these techniques could be utilized to reduce the time spent in the system by 18%.

Instead of addressing the imbalance in workforce capacity, a few studies focused on managing the capacity of other resources at the dining facilities, such as the food stations. Ma et al. (2012) investigated the impact of adding food stations on customer wait times at a CDS location. Their simulation analysis indicated that investing in a new buffet and ice-cream station can reduce the waiting time by 5 min and increase their annual revenue by \$18,000. Li, Shuyuan, Jihan, and Hong (2014) studied the effect of managing the capacity of existing food stations on wait time at a CDS. The authors conducted a detailed process flow analysis and recommended a layout change to shorten the distance traveled at the facility by 33%, which, in turn, reduced the waiting time by 39%.

2.2. Role of queue management on service responsiveness

There is limited literature that has examined the influence of queue management on service quality at dining facilities (Chan and Chan, 2008; Dharmawirya and Adi, 2011; Dixon, Kimes, & Verma, 2009; Ramasamy, Haw, & Chua, 2018). Some studies suggested that the management of queues using technological innovations, such as online-

based ordering and self-service kiosks, can reduce long customer lines (Dixon et al., 2009; Ramasamy et al., 2018). In particular, a cloud-based method to automatically manage the queues of casual dining facilities was explored by Ramasamy et al. (2018), where the customers were allowed to reserve their spot in the queue without being present at the facility. They posited that the proposed system would enable the customers to be served within 10 min of entering the facility. Besides reducing the customer wait time, queue management technologies can also improve their overall experience (Dixon et al., 2009). On the other hand, few studies recommended non-technological strategies to tackle long queues (Chan & Chan, 2008; Dharmawirya and Adi, 2011). For instance, Chan and Chan (2008) suggested the practice of allowing customers to pre-order food while waiting to be seated, as it leads to a responsive service in the subsequent stages of dining. Aside from the dining industry, queue management methodologies have also been adopted to mitigate waiting time in other service sectors, such as hospital emergency departments (Cochran & Roche, 2009; Griffiths, Price-Lloyd, Smithies, & Williams, 2005), call center (Chassioti & Worthington, 2004), banking facilities (Odirichukwu et al., 2013), and logistics (Rajendran, Srinivas, & Saha, 2015; Srinivas, Saha, Yoon, Khasawneh, & Srihari, 2012).

2.3. Research gaps and hypotheses

While prior studies have focused on improving service responsiveness at dining services, to the best of the authors' knowledge, no other work specifically addresses the challenge of managing the workforce in a facility with multiple MTO and pre-prepared food stations under demand uncertainty. Further, most of the previous studies focused on adding capacity to reduce customer waiting time. However, this strategy may not be feasible for many CDS facilities due to resource unavailability or budget restrictions. Based on the findings from existing literature, it is evident that capacity and queue management improve customer waiting times in the hospitality industry, especially dining services. Hence, in this study, the authors derive and test the following three hypotheses for the CDS location under consideration.

H1. Reorganizing the workforce is positively associated with a reduction in the customer waiting time.

H2. Incorporating effective queue management strategy mitigates the delay in service

H3. The combination of queue management and capacity reallocation has a more significant impact on wait time reduction than implementing only one of these two strategies

3. Methodology

Previous research has established the effectiveness of both six sigma and discrete event simulation (DES) methods for improving responsiveness in service and manufacturing industry (Ciulla et al., 2018; Hynes et al., 2019; Lynch, Jamieson, Roussin, & Bae, 2018; Mourtzis, Papakostas, Mavrikios, Makris, & Alexopoulos, 2015; Rajendran et al., 2015; Smith & Srinivas, 2019; Rajendran & Harper, 2020). Six sigma methods provide a structured approach for identifying the root causes of a problem and improving the system performance, while the DES-based approach facilitates virtual modeling and evaluation of any user-defined scenario in an economical and timely manner. To leverage the merits of both these approaches, this paper integrates the six sigma framework that constitutes five phases - Define, Measure, Analyze, Improve, and Control (DMAIC), with DES approach to understand the process, design potential alternatives and evaluate their performance systematically as shown in Fig. 1.

The *Define* phase consists of understanding the current CDS operations and identifying the challenges encountered through stakeholder interaction. Additionally, it involves establishing the desired solution

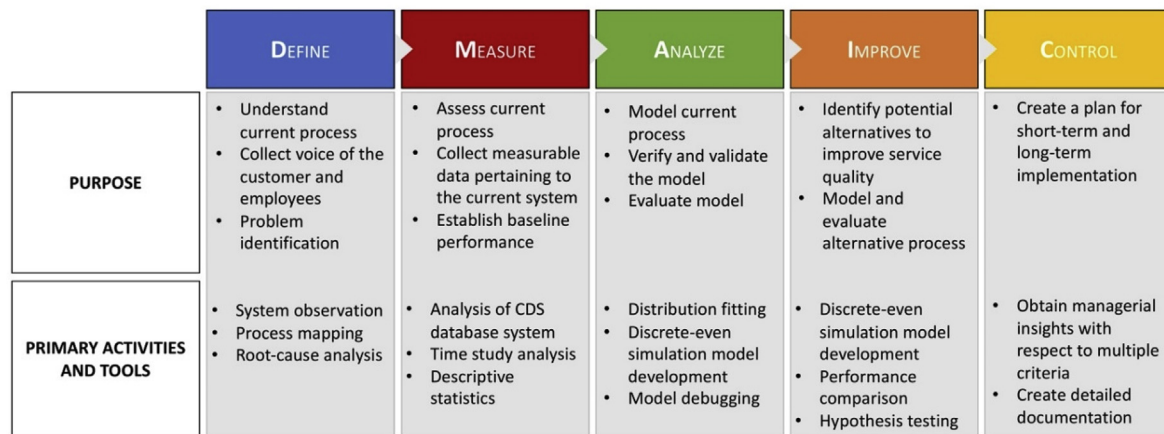


Fig. 1. Overview of simulation-integrated DMAIC framework.

for overcoming the problem. The *Measure* phase comprises of data collection and analyses to assess the current process and establish the baseline performance. Existing resources, such as a campus dining database system (CDDBS), can be leveraged to extract available data, and those that are unavailable in the database can be obtained by conducting a time study at the site. In the *Analyze* phase, descriptive statistics are used to describe the current state of the system by processing the collected data and fitting it to a theoretical distribution. Further, information on the arrival time distribution of students and service time distribution of workers at each station is used to develop a DES model and virtually replicate the current operation of the CDS. In the *Improve* step, potential strategies/configurations for improvement are developed and evaluated logically and systematically using DES. Also, a thorough cost analysis is often performed to help prioritize system improvements. Finally, in the *Control* phase, the new processes and methods are evaluated with respect to multiple criteria to provide managerial insights for sustained short-term and long-term implementation.

4. Case study

The simulation-integrated six sigma framework discussed in Section 3 requires an in-depth understanding of the current operations along with measurable data. Therefore, to demonstrate the feasibility of the integrated methodology, a real-life case study of a CDS located in mid-Missouri, USA, is considered. While the authors study a specific CDS, the proposed methods and alternatives are adaptable to any dining services characterized by an all-you-can-eat style service with MTO stations experiencing high waiting time.

4.1. System observation and process description

The CDS location under study experiences an inflow of approximately 500,000 students every year. The facility is open Monday through Friday and operates in two shifts, namely, lunch (10:30 a.m. to 2:30 p.m.) and dinner (4 p.m.–8 p.m.). It has four MTO stations (Daily Themes, Deli, On-Stage, Grill), three pre-prepared meal stations (Pizza, Salad, Dessert) and, one beverage station. The Pizza, Salad, and Dessert stations are self-serve. On the other hand, Deli and On-Stage stations each have a single chef, while Daily Theme and Grill stations each have two employees. Moreover, one of the two workers at the Grill station and the server at On-Stage station can simultaneously handle four customers. A customer pays a fixed price of \$11 and can have unlimited food at one or more stations. Fig. 2 shows the layout of the CDS facility under study.

The typical customer flow at the CDS is shown in Fig. 3. Upon arrival, the student swipes his/her ID card and pays a fixed price for the

food. The student then chooses one of the eight available stations. If it is an MTO station, then the student places a customized order and waits for it to be prepared before proceeding to the next station or dining table. If there is a queue at the MTO station, then the student must wait before placing an order. On the other hand, if the student chooses a pre-cooked meal station, he/she can typically take the food without having to wait. After consuming the food, the student may choose to have more food by revisiting the food stations or may leave the facility.

The authors interacted with the managers, staff, and customers (or students) at the CDS to understand their concerns. The common issue among students was high waiting time, while the employees raised concerns about long queues and improper resource allocation. Besides, the managers also highlighted various issues pertaining to planning, such as an unpredictable surge in demand. Thus, average customer waiting time is identified to be the critical benchmark of service quality that needs to be improved. Fig. 4 summarizes the primary and secondary causes responsible for high waiting times at the CDS facility under consideration.

4.2. Data collection and visualization

Some data required for this research is extracted from the CDDBS, which includes the arrival time of 180,820 customers and the sales at each station for 16 weeks from August 2018 to November 2018. The data contains 528 missing values and 756 inconsistent values (e.g., negative arrivals, very high sales value), which are discarded during the preliminary analysis. Finally, the cleansed data is analyzed to establish the parameters of the simulation model. A Kruskal-Wallis test indicates no significant difference in the hourly arrivals for different days of the week (p -value = 0.45), thereby allowing them to be combined for distribution fitting. However, the number of arrivals between each hour is significantly different (p -value < 0.05), and a Poisson distribution with a time-varying mean is identified to be a good fit based on the chi-squared test. Fig. 5 shows the average number of hourly arrivals at the CDS. It is observed that an average of 810 and 930 students visit the CDS during the lunch (L) and dinner (D) shifts, respectively. Based on the sales data extracted from the CDDBS and the observations of the system, the proportion of students visiting each station during the lunch and dinner shifts are found to be similar. For both the shifts, Grill station had the highest proportion of student visits (L = 36%, D = 39%) closely followed by the other three MTO stations – On-stage (L = 34%, D = 36%), Daily Themes (L = 31%, D = 33%), and Deli (L = 30%, D = 31%). Likewise, among the pre-cooked stations, the Salad bar has the highest proportion of customer visits (L = 29%, D = 27%), followed by Dessert (L = 18%, D = 20%) and Pizza (L = 17%, D = 16%) stations. Thus, the ranking of the seven buffet stations based on their popularity (i.e., the proportion of student visits) is identical for both the

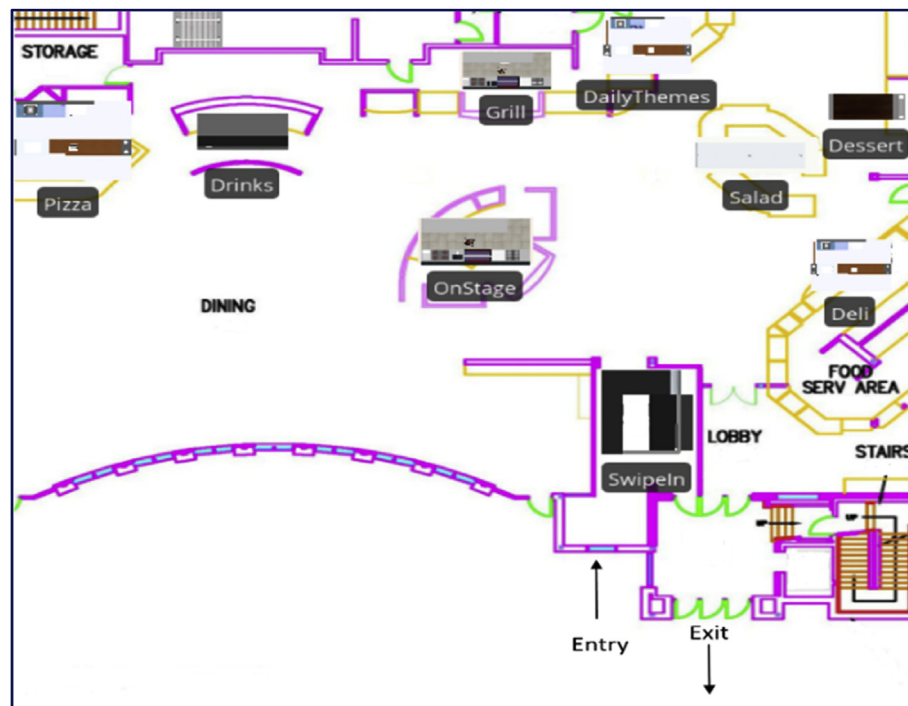


Fig. 2. Layout of the CDS facility under study.

shifts.

The meal preparation time of each order is not recorded in the CDDBS. Therefore, the authors conducted a time study to determine the service time at each station and recorded the meal order and ready time for each customer. The analysis of the service time data indicates low uncertainty at the MTO and pre-cooked stations. Furthermore, the findings are confirmed by the subject matter experts at the facility. Therefore, to incorporate realistic variation, service times (in minutes) are approximated as a uniform distribution over the intervals (2.5, 3) for Grill station, (2.5, 3.5) for On-Stage station, (1, 1.5) for Deli station, (0.5, 1) for Daily Themes, (0.33, 0.50) for Pizza station and (0.83, 0.88) for Drinks station.

4.3. Model development and evaluation of the current process

4.3.1. Discrete event simulation model development

The current operation (Baseline model) of the CDS facility under study is designed and developed in Simio® simulation software by incorporating the system settings, the sequence of events, and process

flow discussed in Section 4.1. The arrival and the service time distributions estimated in Section 4.2 are used as inputs in the simulation model. In order to develop a tractable simulation model, the following simplifying assumptions are considered.

- CDS has sufficient capacity to accommodate the incoming demand.
- There is no restriction on the queue length at the MTO stations.
- In case of multiple servers at a station, they have the same ability and perform tasks at the same rate.
- Customers are catered on a first-come, first-served basis at the MTO stations.

The CDS is modeled as a terminating system as it has fixed operating hours for lunch and dinner. Besides, only the dinner shift (4:00 p.m.–8:00 p.m.) is chosen to be modeled as its average demand is higher than the lunch shift. Since the proportion of student visits is similar for both the shifts, it is expected that any strategy that improves the wait time during the dinner shift would also enhance the lunch shift performance. The DES model is run for 50 replications with a warm-up

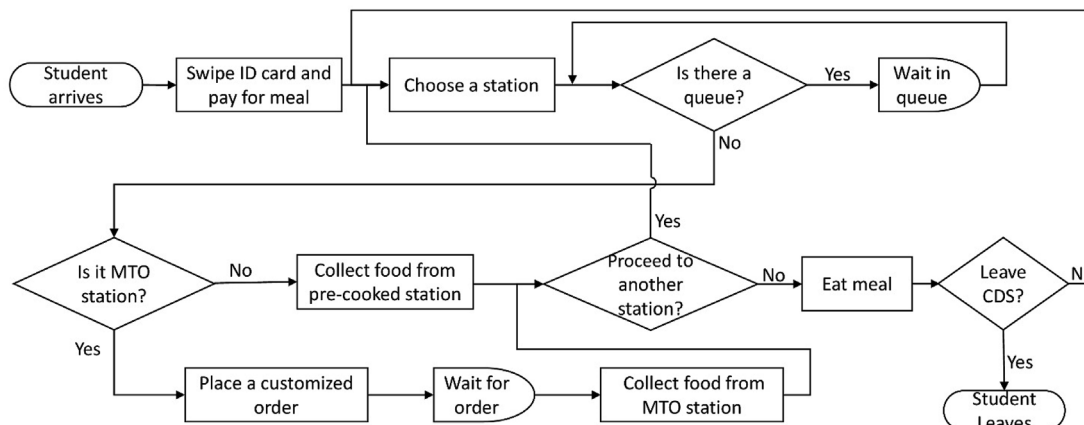


Fig. 3. A typical flow of a student arriving at CDS.

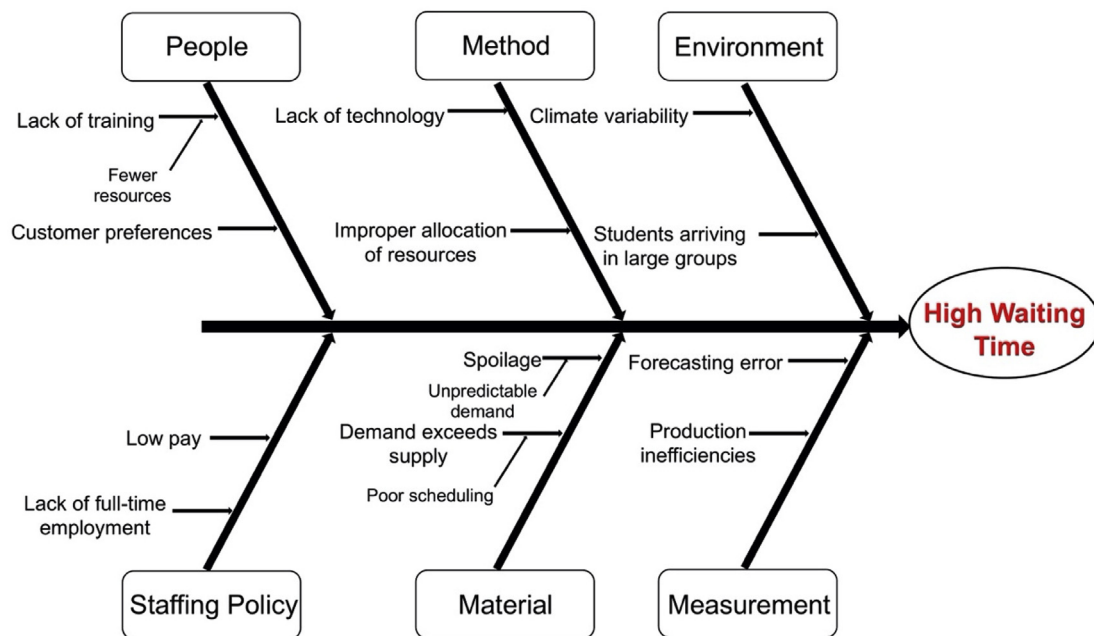


Fig. 4. Root cause analysis of high waiting time at the CDS under study.

period of 1 h.

4.3.2. Model verification, validation, and results

The Baseline model is verified using various debugging tools available within Simio®. The flow of entity and server functionalities in the simulation model are monitored at various instances during the run using the 'Breakpoint' function, which pauses the simulation when a user-defined condition is met. Besides, the 'Model Trace' feature is also used to extract thorough information on each entity's attributes from entry to exit. Upon verification, the model is validated based on two measures, namely, the number of customer arrivals and average waiting time in the system. The actual system has 400 customers arriving during peak hours and experiences an average waiting time of 20 min. The Baseline model (or simulation of the current system) results in 412 arrivals and 23.24 min of average waiting time. A *t*-test shows that the validation measures of the baseline model are not statistically different from the current system, thereby validating the simulation model. The results of the simulation model indicate the Grill and On-Stage stations to be the primary cause for the long waiting time (i.e., bottleneck).

Besides, the following two issues are also evident in the simulation model of the current system.

- **Imbalanced Worker Utilization at Grill Station:** Even though the Grill station has two employees, only one staff is preparing the MTO food, while the other worker is primarily serving the grilled meals. Therefore, the staff preparing the meals is overburdened, while the staff serving it is underutilized.
- **Improper Queue Management:** The current process at the MTO stations, Grill and On-Stage, requires the students to place their order and wait in line while the server prepares their food. This prevents the student from leaving the queue to another station. Moreover, this also blocks the subsequent customer from placing an order, thereby leading to long waiting times during peak hours.

While adding additional workers would improve the current situation, it is not cost-effective and may not be feasible due to resource shortages. Therefore, the authors propose alternatives to address the aforementioned vital issues through capacity reallocation and queue

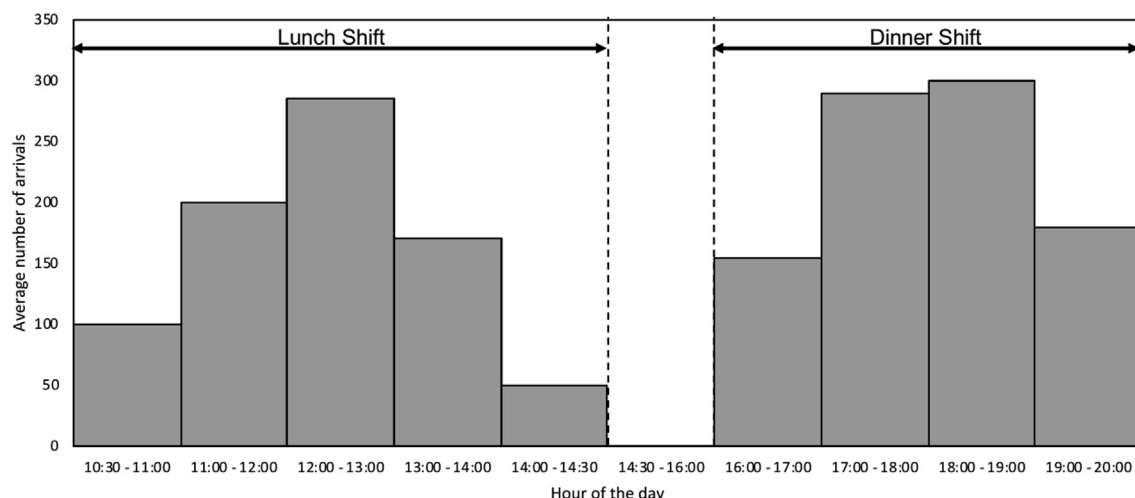


Fig. 5. Hourly arrival rate of customers at the CDS facility for a typical day.

management.

5. Design and evaluation of alternatives

5.1. Proposed alternatives

- **Scenario 1: Impact of workforce reallocation** – One of the two servers at the Grill station is underutilized and, therefore, re-allocated to be shared between the two bottleneck stations - Grill and On-Stage. Based on sales data extracted from CDDBS and the facility visits, it is observed that students did not join the queue if its length at the Grill and On-Stage stations is approximately over 20 and 15, respectively. Therefore, it is proposed to allocate the shared worker to the Grill station if and only if, the number of students in the queue is more than 20 for Grill and less than 15 for On-Stage. Otherwise, the resource is commissioned to On-Stage station due to its relatively higher service time.
- **Scenario 2: Impact of check-sheet-based queue management** – The second alternative evaluates the effect of untethering customers from the queue by implementing reusable check sheets at the two MTO stations – Grill and On-Stage. Rather than waiting in queue for their food to be prepared, this scenario would allow the students to provide instructions for the MTO food on the check sheet and collect their order once it is ready. While the meal is being prepared, the customer can visit and consume food from other stations.
- **Scenario 3: Impact of workforce reallocation and check-sheet-based queue management** – This scenario assesses the influence of using a shared resource between On-Stage and Grill while adopting reusable check sheets at these stations. Since the third alternative involves a combination of the first two scenarios, it seeks to measure the combined effects of capacity reallocation and queue management.
- **Scenario 4: Impact of technology-based queue management system** – A potential shortcoming of the second alternative is that students are bound to make multiple visits to the corresponding station(s) to check if their food is ready. To avoid such a situation, the fourth alternative explores the use of Arduino-Uno Interface – a queue management system where customers place an order using a digital kiosk and enter their phone number to receive an automated text message when the food has been prepared. Previous studies have shown that technologically competent people, such as the current generation of college students, are more likely to accept and engage with Arduino-Uno Interface (Govindasamy, 2001; Pérez-Aróstegui, Bustinza-Sánchez, & Barrales-Molina, 2015).
- **Scenario 5: Impact of capacity reallocation and technology-based queue management** – The fifth alternative evaluates the combined impact of capacity reallocation (Scenario 1) and remote sensing queue management system (Scenario 4). In other words, this scenario assesses the effect of using a shared resource and integrating the Arduino-Uno interface for queue management.

5.2. Results of proposed alternatives

To ensure a fair comparison, the proposed alternatives are evaluated by running the simulation model for the same experimental settings as the Baseline scenario. It is evident from Table 1 that all the proposed alternatives outperform the existing operation as it yields a lower average waiting time. The incremental cost required to implement the proposed alternatives is also estimated, as shown in Table 1. Scenario 1 requires the least investment as it only incurs a fixed staff reallocation cost of \$200. Although assigning an existing server to two stations does not incur any additional cost, it is projected that the CDS would be subjected to a one-time labor charge of \$200 to relocate and set up the equipment at these stations. For Scenario 2, it is anticipated that the management would need to invest in 140 check sheets every year based on the total queue length at the MTO stations. This will incur an annual

Table 1

Average waiting time under different scenarios.

Alternatives	Waiting Time (mins)	Incremental Fixed Cost (\$)	Incremental Annual Cost (\$/year)
Baseline	23.24	–	–
Scenario 1	16.45	200	0
Scenario 2	14.65	0	420
Scenario 3	12.74	200	420
Scenario 4	12.43	8200	3100
Scenario 5	11.41	8400	3100

Table 2

Pairwise *t*-test results for the proposed hypothesis.

Hypothesis	Scenarios Compared	$\overline{WT}_{diff}$ (SE)	Result
H1	Baseline – Scenario 1	6.79 (0.29)***	Supported
H2	Baseline – Scenario 2	8.59 (0.27) ***	Supported
	Baseline – Scenario 4	10.81 (0.28)***	
H3	Scenario 1 – Scenario 3	3.71 (0.21)***	Supported
	Scenario 2 – Scenario 3	1.91 (0.19)***	
	Scenario 1 – Scenario 5	5.04 (0.19)***	
	Scenario 4 – Scenario 5	1.02 (0.18)***	

Note: $\overline{WT}_{diff}$ = Mean difference in waiting time; SE = Standard Error.

***p < 0.001; **p < 0.01; *p < 0.05.

cost of \$420 (\$3 per check sheet x 140 check sheets per year). If the management instead elects a more sustainable solution such as the Arduino-Uno-based queue management system, then it will be subjected to a one-time investment of \$8200 ([(\$3700 per Hardware Equipment + \$400 installation cost per kiosk] x 2 MTO stations) and variable cost of \$3100/year (\$600 for software + \$2500 for technical support). Scenario 3 will incur the combined cost of Scenarios 1 and 2, while Scenario 5 will endure the costs of Scenarios 1 and 4.

To test the proposed hypotheses, multiple pairwise *t*-tests are conducted using SPSS 26.0 and their results are summarized in Table 2. The relationships in Hypothesis 1 is examined based on the difference in mean waiting time between alternatives that reallocates the workforce (Scenario 1) and maintains an existing assignment (Baseline model). Their mean difference is observed to be statistically significant ($\overline{WT}_{diff}$ = 6.79; SE = 0.29, *p* < 0.001), supporting Hypotheses 1. Similarly, the influence of queue management strategies on responsiveness is obtained by comparing the waiting time of the Baseline scenario with an alternative that manages queue with a reusable check sheet (Scenario 2) and Arduino-Uno interface (Scenario 4), respectively. The results demonstrate that the reduction in wait time based on the two queue management strategies is significantly better than the Baseline model, thereby supporting Hypothesis 2. Finally, the adoption of both capacity and queue management strategies exhibits significant improvement over their individual implementation, therefore supporting Hypothesis 3. For instance, Alternative 3 (which combines Scenarios 1 and 2), has a considerably lower waiting time than Scenario 1 ($\overline{WT}_{diff}$ = 3.71; SE = 0.21, *p* < 0.001) and Scenario 2 ($\overline{WT}_{diff}$ = 1.91; SE = 0.19, *p* < 0.001). Likewise, Scenario 5 yields better performance than Scenario 1 ($\overline{WT}_{diff}$ = 5.04; SE = 0.19, *p* < 0.001), as well as Scenario 4 ($\overline{WT}_{diff}$ = 1.02; SE = 0.18, *p* < 0.001).

5.3. Impact of increased arrivals

In the initial analysis, the data from the CDDBS is used to estimate the average hourly arrival rate. Since the student enrollment is expected to increase in the forthcoming years, the demand for CDS is also anticipated to grow. In this section, the impact of increased customer arrivals at CDS is examined, where hourly arrival rates are increased by 5%, 10%, 15%, and 20%.

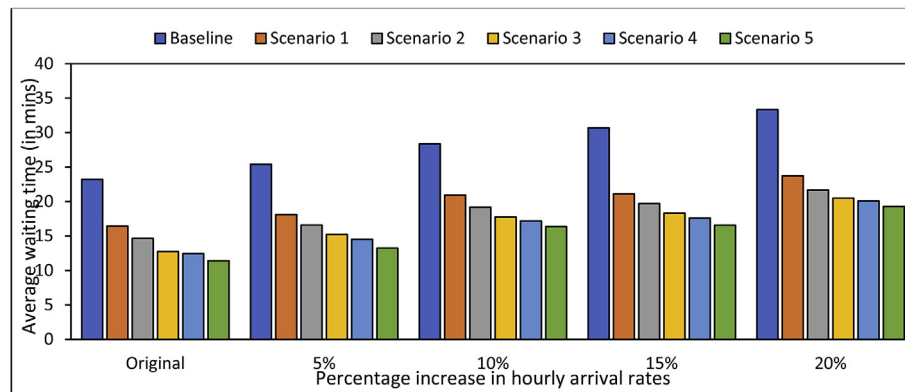


Fig. 6. Impact of increased arrivals on average waiting time.

Fig. 6 shows the impact of increasing arrivals on average waiting time. As expected, the waiting time increases with a rise in demand for all the scenarios under consideration. For example, the average waiting time increases by 47% for the Baseline model when the demand grows by 20%. However, all the proposed alternatives consistently outperform the current process even when the number of arrivals is increased. Moreover, consistent with earlier findings, Scenario 5 yields the lowest waiting time in all the cases.

6. Discussion

6.1. Theoretical implications

This research adds to the knowledge of improving responsiveness with constrained resources at campus dining facilities. The current study examines the independent and combined effects of capacity reallocation and queue management on customer waiting time at a CDS. The extant literature on managing service quality in the hospitality industry has identified capacity and queue management strategies to play a substantial role in reducing customer wait times (Bruneel & Wittevrongel, 2017; Curin et al., 2005; Iqbal et al., 2012; Wijweickrama, 2006). The present findings also concur with these conclusions in the literature.

The scenarios focusing on efficiently shifting existing workforce to stations experiencing high demand has shown to improve the responsiveness, and also verify the previous findings in the literature (Ghaleb et al., 2015). An important take-away is that the reallocation must be managed by considering the multi-station multi-queue setting of the CDS to achieve the desired outcome. Otherwise, shifting resources to reduce waiting time at one station may lead to longer queues at other stations (Dickson, Ford, & Laval, 2005). Besides, this study demonstrates that the service responsiveness can be significantly improved by allowing the customers to disengage a queue upon placing an MTO order using reusable check sheets or Arduino-Uno Interface. Therefore, the customer's wait in the queue for the MTO food is virtual, since they are allowed to visit other food stations and eat, while their MTO order is being prepared. This conclusion aligns with the finding of previous works on improving hospitality services, in which customer wait time is reduced after reserving their place in line for services, instead of being delayed due to long queues (Chang & Yang, 2008; Kokkinou & Cranage, 2013).

This study also reveals that the integration of queue management and capacity reallocation strategies can achieve significantly better improvement in waiting time than implementing only one of these two strategies. Although existing literature on improving hospitality services did not report similar findings, the results can be supported by the conclusions of Srinivas et al. (2012) and Rajendran et al. (2015) in the logistics industry. This paper also provides a new perspective on researching service quality in the foodservice industry by integrating a

simulation-based approach with a data-driven quality improvement framework. Similar hybrid approaches have shown to improve operational performance in healthcare and transportation sectors as well (e.g., Hussein, Abdelmaguid, Tawfik, & Ahmed, 2017; Smith & Srinivas, 2019; Venkatadri, Raghavan, Kesavakumaran, Lam, & Srihari, 2011). Also, the results of the current study deepen the support for the use of six-sigma tools such as value stream mapping (Hynes et al., 2019), process flow maps (Ciulla et al., 2018) to reduce the waiting time in the service industry.

6.2. Managerial implications

The results of this study provide several implications for dining facility managers/operators. First, the study indicates that the existing labor pool can be effectively managed to provide faster service to diners. Thus, even if the managers are unable to expand the workforce due to budget restrictions or resource unavailability, they could effectively shift and share appropriate resources among stations to curtail the waiting time. Besides, the study provides different possibilities to manage capacity and customer queues. While all the alternatives have the potential to improve the responsiveness substantially, managers should consider other factors (such as the cost of adopting a new process and time required to implement a change) before pursuing any recommendation.

An alternative that requires tremendous capital investment for a marginal increase in performance is undesirable if there is a significant payback period (Madanoglu, Erdem, & Gursoy, 2008). A cash-strapped dining facility should consider low-cost alternatives such as server reallocation and paper-based queue management system (e.g., Scenarios 1, 2, or 3) for enhancing the responsiveness. On the other hand, a CDS with commensurable liquid capital can adopt a technology-based queue management system (Scenario 4) in addition to repositioning their workforce (Scenario 5). Besides refining the diner's experience, a technology-based system would also provide a sustainable option as it eliminates the need for paper check-sheets. With sustainability concerns on the rise, customers are avoiding products and services that directly affect the environment (Hu, Parsa, & Self, 2010; Sarmiento & El Hanandeh, 2018). Hence, these alternatives would not repel environmentally conscious customers.

Aside from implementation cost, the decision-makers must also contemplate about the time required to enforce the change. This is mainly because the layout and ambiance of a facility have shown to have a positive disconfirmation on customer satisfaction (Ryu & Han, 2011). Therefore, any systemic change that requires substantial renovation time may not provide a pleasant ambiance for dining, which, in turn, impacts the customer experience. Thus, if the CDS manager is required to institute a change during the academic term (e.g., Fall or Spring semesters), then the student influx is high, and the renovations must be completed promptly to avoid any inconveniences. Under such

circumstances, using alternatives that require simple process change (e.g., check-sheets for handling long queues and/or worker reallocation) is recommended as it can be implemented quickly. However, if the change is planned during periods of low demand (e.g., summer break), then alternatives yielding the most significant improvement in service quality can be adopted without being concerned about its implementation time. In particular, the installation of new hardware such as self-serving kiosks and Arduino-Uno interface is suggested, as it significantly reduces waiting time but requires longer implementation time due to setup, installation, and integration with existing database systems.

Based on the needs and capital availability, the CDS can adopt one of the proposed solutions to improve service quality for the lunch shift as well. Nevertheless, it is beneficial for the management to implement the same strategy for both the shifts. This is because the ranking of the alternatives with respect to waiting time reduction is expected to be the same for lunch and dinner shifts since the proportion of students visiting each station is similar for both the shifts. Besides, if the management invests in the resources required for implementing an alternative in the dinner shift, then adopting the same alternative will not require any substantial additional expenditure.

6.3. Limitations and future work

This study has some limitations and also provides directions for future research. First, this paper investigates the impact of the proposed alternatives using simulation modeling for a single CDS. This limits the ability to generalize the effectiveness of the alternatives evaluated. Hence, future works could replicate the impact of the proposed approach at other similar campus dining facilities to strengthen the findings of this research. Besides, the present study is solely focused on a campus dining establishment. This provides an opportunity for researchers to extend the study to other facilities such as fast-food and fine dining restaurants. Third, the focus of this study is to improve one aspect of service quality, namely, responsiveness. However, there might be other influencing factors, such as food quality and employee training, that might affect service level and customer satisfaction. Further studies could examine the role of all these factors on different dimensions of service quality. Fourth, the present study does not optimize the placement of the self-service kiosks in the CDS, but future studies could examine it to provide better customer experience. Finally, this research requires customers to be physically present to place an order. To avoid congestion at the facility and provide more flexibility to the customer, the impact of allowing online orders could be considered as a potential scope of improvement.

7. Conclusions

Long waits at campus dining service have a detrimental impact on customer satisfaction. Most of the previous works focused on hiring additional workers to provide prompt service. However, capacity expansion may not be feasible due to budget restrictions or the lack of a skilled workforce. Unlike prior studies, this research seeks to improve service responsiveness at campus dining service by managing the existing workforce and customer queues. The findings indicate that the reallocation of underutilized servers and untethering customers from long queues can ameliorate operational inefficiencies and reduce customer waiting time. Besides, the current study has offered a systematic framework for virtually evaluating the impacts of potential changes to dining operations by integrating six-sigma principles and simulation modeling. In addition, this paper also contributes to the dearth of existing literature on minimizing waiting time at dining establishments. Based on this study's findings, managers can adopt prudent strategies to curtail long queues and excessive waits for the customers and enhance their dining experience.

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Declaration of competing interest

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Appendix A. Supplementary data

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References

- Adenso-Díaz, B., González-Torre, P., & García, V. (2002). A capacity management model in service industries. *International Journal of Service Industry Management*, 13(3–4), 286–302.
- Brandmeir, K., & Baloglu, S. (2005). Linking employee turnover to casino restaurant performance: A cross-sectional and time-lagged correlation analysis. *Journal of Foodservice Business Research*, 7(2), 25–39.
- Bruneel, H., & Wittevrongel, S. (2017). Analysis of a discrete-time single-server queue with an occasional extra server. *Performance Evaluation*, 116, 119–142.
- Chan, W. W., & Chan, L. C. (2008). Revenue management strategies under the lunar-solar calendar: Evidence of Chinese restaurant operations. *International Journal of Hospitality Management*, 27(3), 381–390.
- Chang, H. L., & Yang, C. H. (2008). Do airline self-service check-in kiosks meet the needs of passengers? *Tourism Management*, 29(5), 980–993.
- Chassioti, E., & Worthington, D. J. (2004). A new model for call centre queue management. *Journal of the Operational Research Society*, 55(12), 1352–1357.
- Ciulla, T. A., Tatikonda, M. V., ElMaraghi, Y. A., Hussain, R. M., Hill, A. L., Clary, J. M., et al. (2018). *Lean six sigma techniques to improve ophthalmology clinic efficiency*, Vol. 38, Philadelphia, Pa: Retina1688–1698 9.
- Cochran, J. K., & Roche, K. T. (2009). A multi-class queueing network analysis methodology for improving hospital emergency department performance. *Computers & Operations Research*, 36(5), 1497–1512.
- Curin, S. A., Vosko, J. S., Chan, E. W., & Tsimhoni, O. (2005). *Reducing service time at a busy fast food restaurant on campus. Proceedings - winter simulation conference*, Vol. 2005, 2628–2635.
- Dharmawirya, M., Adi, E., & Case Study for Restaurant Queuing Model (January 1, 2011). *2011 international conference on management and artificial intelligence* 52–55.
- Dickson, D., Ford, R. C., & Laval, B. (2005). Managing real and virtual waits in hospitality and service organizations. *Cornell Hotel and Restaurant Administration Quarterly*, 46(1), 52–68.
- DiPietro, R. B., Parsa, H. G., & Gregory, A. (2011). Restaurant QSC inspections and financial performance: An empirical investigation. *International Journal of Contemporary Hospitality Management*, 23(7), 982–999.
- Dixon, M., Kimes, S. E., & Verma, R. (2009). Customer preferences for restaurant technology innovations. *Cornell Hospitality Report*, 9(7), 6–16.
- El-Said, O. A., & Fathy, E. A. (2015). Assessing university students' satisfaction with on-campus cafeteria services. *Tourism Management Perspectives*, 16, 318–324.
- Fraginière, E., Gondzio, J., & Yang, X. (2010). Operations risk management by optimally planning the qualified workforce capacity. *European Journal of Operational Research*, 202(2), 518–527.
- Ghaleb, M. A., Suryahatmaja, U. S., & Alharkan, I. M. (2015). Modeling and simulation of queueing systems using arena software: A case study. *IEOM 2015 - 5th international conference on industrial engineering and operations management, proceeding* (pp. 1–7). Institute of Electrical and Electronics Engineers.
- Ghouri, A. M., Khan, N. U., Malik, M. A., & Razzaq, A. (2011). Marketing practices and their effects on firm's performance: Findings from small and medium sized catering and restaurants in Karachi. *International Journal of Business and Management*, 6(5), 251–259.
- Govindasamy, T. (2001). Successful implementation of e-learning: Pedagogical considerations. *The Internet and Higher Education*, 4(3–4), 287–299.
- Griffiths, J. D., Price-Lloyd, N., Smithies, M., & Williams, J. E. (2005). Modelling the requirement for supplementary nurses in an intensive care unit. *Journal of the Operational Research Society*, 56(2), 126–133.

- Gupta, S., McLaughlin, E., & Gomez, M. (2007). Guest satisfaction and restaurant performance. *Cornell Hotel and Restaurant Administration Quarterly*, 48(3), 284–298.
- Gurgol, D. S. (2019). A student's journey to career readiness: Maximizing on-campus employment. *M.A. in Higher Education Leadership: Action Research Projects*, 29.
- Hu, H. H., Parsa, H. G., & Self, J. (2010). The dynamics of green restaurant patronage. *Cornell Hospitality Quarterly*, 51(3), 344–362.
- Hussein, N. A., Abdelmaguid, T. F., Tawfik, B. S., & Ahmed, N. G. S. (2017). Mitigating overcrowding in emergency departments using six sigma and simulation: A case study in Egypt. *Operations Research for Health Care*, 15, 1–12.
- Hynes, J. P., Murray, A. S., Murray, O. M., Eustace, S. K., Gilchrist, S., Dolan, A., et al. (2019). Use of Lean Six Sigma methodology shows reduction of inpatient waiting time for peripherally inserted central catheter placement. *Clinical Radiology*, 74(9), 733.e5–733.e9.
- Iqbal, Q., Whitman, L. E., & Malzahn, D. (2012). Reducing customer wait time at a fast food restaurant on campus. *Journal of Foodservice Business Research*, 15(4), 319–334.
- Joung, H. W., Lee, D. S., Kim, H. S., & Huffman, L. (2014). Evaluation of the on-campus dining services using importance-performance analysis. *Journal of Foodservice Business Research*, 17(2), 136–146.
- Kim, H. S., Jang, S. C., & Shawn (2020). Does a minimum wage increase endanger restaurant jobs? Examining the role of franchising. *International Journal of Hospitality Management*, 84.
- Klassen, K. J., Trybus, E., & Kumar, A. (2005). Planning food services for a campus setting. *International Journal of Hospitality Management*, 24(4), 579–609.
- Kokkinou, A., & Cranage, D. A. (2013). Using self-service technology to reduce customer waiting times. *International Journal of Hospitality Management*, 33(1), 435–445.
- Lee, S., Singal, M., & Kang, K. H. (2013). The corporate social responsibility–financial performance link in the US restaurant industry: Do economic conditions matter? *International Journal of Hospitality Management*, 32, 2–10.
- Li, H., Shuyuan, J., Jihan, J. C., & Hong, C. (2014). Innovative applications of IE technological in the service sector: The case study- A case of panzhuhua university canteen service system. *Journal of Applied Science and Engineering Innovation*, 1(5), 338–344.
- Lynch, H. M., Jamieson, K., Roussin, C., & Bae, D. (2018). Increasing healthcare value through simulation: Cost savings from reductions in cast saw injuries after simulation-based education of orthopaedic trainees. *Pediatrics*, 141, 626–626.
- Ma, X., Beeker, A., McCloskey, M., Moldawer, D., & Williams, E. (2012). Increasing efficiency of large university dining hall usage. *Proceedings of the International Workshop on Applied Modeling and Simulation*, 14–17.
- Madanoglu, M., Erdem, M., & Gursoy, D. (2008). Risk return and cost of equity of small and large casual-dining restaurants. *International Journal of Hospitality Management*, 27(1), 109–118.
- McCarthy, B. Understaffing creates plastic problem for campus dining services. (2018). Retrieved from <https://move.themaneater.com/stories/move-investigates/move-investigates-understaffing-creates-plastic-problem-for-campus-dining-services> Accessed date: July 30, 2019.
- Mourtzis, D., Papakostas, N., Mavrikios, D., Makris, S., & Alexopoulos, K. (2015). The role of simulation in digital manufacturing: Applications and outlook. *International Journal of Computer Integrated Manufacturing*, 28(1), 3–24.
- Odirichukwu, J. C., Lekara, T., & Odii, J. N. (2013). Banking queue system in Nigeria. *Computing, Information Systems, Development Informatics and Allied Research Journal*, 5(1), 98–105.
- Parasuraman, A., Zeithaml, V. A., & Berry, L. L. (1988). SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality. *Journal of Retailing*, 64(1), 12–40.
- Pérez-Aróstegui, M., Bustinza-Sánchez, F., & Barrales-Molina, V. (2015). Exploring the relationship between information technology competence and quality management. *BRQ Business Research Quarterly*, 18(1), 4–17.
- Pullman, M., & Rodgers, S. (2010). Capacity management for hospitality and tourism: A review of current approaches. *International Journal of Hospitality Management*, 29(1), 177–187.
- Rajendran, S., & Harper, A. (2020). A simulation-based approach to provide insights on Hyperloop network operations. *Transportation Research Interdisciplinary Perspectives*. <http://dx.doi.org/10.1016/j.trip.2020.100092> In press.
- Rajendran, S., Srinivas, S., & Saha, C. (2015). Analysis of operations of port using mathematical and simulation modelling. *International Journal of Logistics Systems and Management*, 20(3), 325–347.
- Ramasamy, R., Haw, S., & Chua, F. (2018). Casual dining restaurant queue management system to optimize decision making in table seating arrangement. *Knowledge management international conference, Malaysia* (pp. 18–22).
- Ryu, K., & Han, H. (2011). New or repeat customers: How does physical environment influence their restaurant experience? *International Journal of Hospitality Management*, 30(3), 599–611.
- Sarmiento, C. V., & El Hanandeh, A. (2018). Customers' perceptions and expectations of environmentally sustainable restaurant and the development of green index: The case of the Gold Coast, Australia. *Sustainable Production and Consumption*, 15, 16–24.
- Smith, D., & Srinivas, S. (2019). A simulation-based evaluation of warehouse check-in strategies for improving inbound logistics operations. *Simulation Modelling Practice and Theory*, 94, 303–320.
- Srinivas, S., Saha, C., Yoon, S., Khasawneh, M., & Srihari, K. (2012). Analysis of campus traffic congestion during move-in days using discrete event simulation model. *Proceedings of the 1st annual world conference of the society for industrial and systems engineering*. Washington, D.C, USA, 16-18 september 2012.
- Tang, T. Q., Zhang, B. T., Zhang, J., & Wang, T. (2019). Statistical analysis and modeling of pedestrian flow in university canteen during peak period. *Physica A: Statistical Mechanics and its Applications*, 521, 29–40.
- Technomic. Students food service demands grow both on and off campus. (2019). Retrieved from <https://www.technomic.com/newsroom/students-foodservice-demands-grow-both-campus-technomic-finds-2019-college> Accessed date: July 25, 2019.
- Velumani, S., Pitchiah, S., & Tang, H. (2017). A study of service wait time and its improvement in cafeteria using discrete event simulation. *International Journal of Research in Engineering and Technology*, 6(10), 30–39.
- Venkatadri, V., Raghavan, V. A., Kesavakumaran, V., Lam, S. S., & Srihari, K. (2011). Simulation based alternatives for overall process improvement at the cardiac catheterization lab. *Simulation Modelling Practice and Theory*, 19(7), 1544–1557.
- Wijewickrama, A. K. A. (2006). Simulation analysis for reducing queues in mixed-patients' outpatient department. *International Journal of Simulation Modelling*, 5(2), 56–68.
- Zhao, Z. (2016). Restaurant queuing management in experience economy era. *DEStech Transactions on Environment, Energy and Earth Science*, 1–5. <https://doi.org/10.12783/dteees/peee2016/3860>.